MILITARY SPECIFICATION

BULLET, CAL .30

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, packaging, the fabrication, manufacture of one type of bullet designated as Bullet, Cal .30.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition Through Cal .50
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert ammunition Components

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein:

DRAWING

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKING DRAWINGS

8595432 - Cartridge, Cal .30 Match, M72
8595434 Bullet

INSPECTION EQUIPMENT DRAWINGS

EL 8658203 - Velocity
EL 8658216 - Accuracy

PUBLICATIONS

SCATP-7.62 - Ammunition Ballistic Acceptance Test Methods, Test Procedures for 7.62MM Cartridge

2.1.3 Order of precedence. In the event of a conflict between this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General The bullet shall comply with the requirements of Drawing (Dwg.) 8595434, referenced specifications and other requirements specified herein.

3.2 Accuracy. The bullet shall be assembled in accordance with Dwg. 8595432 and the average of the mean radii of all targets from the sample cartridges fired at 600 yards shall not be greater than 3.5 inches.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract. (See 4.3).

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3.4 Workmanship. The bullet shall be fabricated in a thorough and workmanlike manner. It shall be free of burrs, sharp edges, foreign matter, and defects not permitted by MIL-STD-636. The cleaning method used shall not be injurious to the bullet, nor shall the bullet be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply are hereby made part of this detail specification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3`

b. Quality conformance Inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

Part Description	Drawing	Quantity
Cartridge, Cal .30 Match, M72	8595432	400
Bullet	8595434	125
Jacket	8595435	125
Slug	8595437	125

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078 and Table II.

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Table I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-B-70445(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Bullet and Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Slug, (Dwg. 8595437)				
	Examination for defects	125	ACC 0 REJ 1	3.1, 3.4	4.4.2.1
	Jacket (Dwg. 8595435)				
	Examination for defects	125	ACC 0 REJ 1	3.1, 3.4	4.4.2.2
	Bullet (Dwg. 8595434)				
	Examination for defects	125	ACC 0 REJ 1	3.1, 3.4	4.4.2.3
	Accuracy	150 1/	2/	3.2	4.5.1 EL 8658203 EL 8658216

NOTES: 1/ Represents 15 targets with 10 cartridges fired at each.
2/ The average of the mean radii of all targets from the sample cartridges shall meet the applicable requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain bullets from one interfix lot number from one manufacturer. Inspection lots shall not exceed 150,000 bullets per lot.

4.4.2 Examination. See MIL-A-48078.

Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70445(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8595437
4.4.2.1	Slug				NEXT HIGHER ASSEMBLY 8595434
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Weight		.25%	3.1	SMTE
MINOR:					
201.	Evidence of poor workmanship		1.5%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70445(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Jacket	1 OF 1	8595435
			NEXT HIGHER ASSEMBLY: 8595434

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101.	Wall thickness		.25%	3.1	SMTE
102.	Variation in wall thickness		.25%	3.1	SMTE
103.	Outside diameter		.25%	3.1	SMTE
104.	Diameter at tip, maximum		.25%	3.1	SMTE
MINOR:					
201.	Evidence of poor workmanship		1.5%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70445(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8595434
4.4.2.3	Bullet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined.				
MAJOR:					
101.	Dent		.25%	3.4	Visual 1/
102.	Scratch		.25%	3.4	Visual 1/
103.	Split bullet jacket		.25%	3.4	Visual 1/
104.	Scaly metal		.25%	3.4	Visual 1/
105.	Upset point		.25%	3.4	Visual 1/
106.	Blunt point		.25%	3.4	Visual 1/
107.	Evidence of poor workmanship		.25%	3.4	Visual
108.	Overall length		.25%	3.1	SMTE
109.	Outside diameter		.25%	3.1	SMTE
110.	Angle of boattail		.25%	3.1	SMTE
111.	Length of boattail		.25%	3.1	SMTE
112.	Radius of ogive		.25%	3.1	SMTE
113.	Diameter at tip, maximum		.25%	3.1	SMTE
114.	Weight		.25%	3.1	SMTE
115.	Accuracy	90 2/	3/	3.2	EL 8658203 EL 8658216

NOTES:

1/ Refer to MIL-STD-636 for visual defect standards. In the event of a conflict between paragraph 4.4.2.3 of this specification and MIL-STD-636 as to defect classification, the classification specified in paragraph 4.4.2.3 shall apply.

2/ Represents 9 targets with 10 cartridges fired at each.

3/ The average of the mean radii of all targets from the sample shall meet the applicable requirements.

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4.4.3 Testing. Testing is described in Table 1 and paragraph 4.4.2.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in Table 1 and paragraphs 4.4.2.1 through 4.4.2.3. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Accuracy. Prior to assembly of this sample, a quantity of 20 cartridges assembled in accordance with Dwg. 8595432, shall be tested for velocity in accordance with Section 18 of SCATP-7.62. The test shall be conducted at $70^{\circ} \pm 2^{\circ}F$. If the average velocity of the quantity is not between 2640 feet per second (fps) $\pm$ 30 fps at 78 feet from the weapon muzzle or if the standard deviation of the velocities is greater than 28 fps, a new charge weight must be established. When a quantity of 20 cartridges complies with these requirements, the accuracy sample shall be assembled in accordance with Dwg. 8595432 using the assessed charge weight. The charge weight and overall length of each assembled cartridge from both the assessment quantity and accuracy sample shall be measured and the values recorded. The accuracy test shall be conducted in accordance with Section 3 of SCATP-7.62 using a range of 300 yards or 600 yards. A range of less than 300 yards shall not be used. When the test is conducted on a 300 yard range, the average of the mean radii of the targets shall be multiplied by the conversion factor of 2.3.

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified, packaging shall be Level B. Maximum and intermediate protection (Level A and B) shall be in accordance with MIL-STD-1169. Minimum protection shall be in accordance with MIL-STD-1188.

5.2 Unitization. Unitization for Levels A and B shall be in accordance with MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Intended use. The bullet covered by this specification is intended for National Match use.

6.2 Ordering data.

a. See MIL-A-48078

b. Responsibility for furnishing brass cases, propellant and primers (See 3.2).

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, ATTN: DRSMC-QAF-I(D) Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as U.S. Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

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